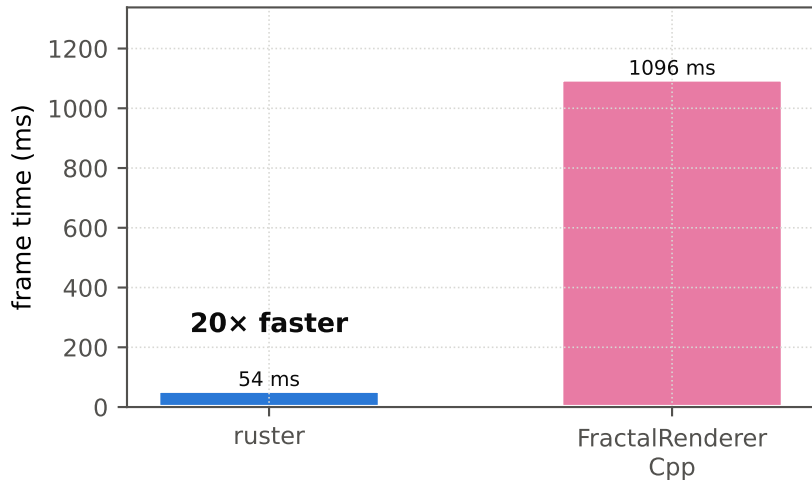
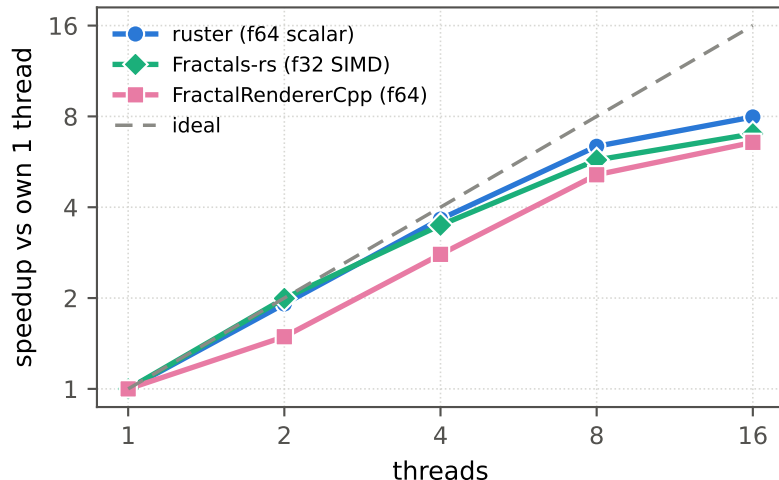


Single-thread f64 scalar (fully controlled comparison)



Parallel scaling (normalised)



LEFT is the only fully controlled cross-project comparison — same fractal, resolution, iteration count, precision, algorithm class and thread count — so it isolates kernel quality: ruster rejects cardioid/period-2/period-3 interiors and detects cycles; the C++ kernel does neither. RIGHT normalises each project against its own single thread, so it compares scaling shape only, not speed. $\triangle$ The two projects' numbers come from different sessions; quote the left panel as 17–20 $\times$ until all three are re-run together (see other-projects/tri-compare).